www.tntextbooks.in

as a foliar spray to plants, induces partial stomatal closure for two weeks or more without any toxic effect. Use of **abscisic acid** highly induces the closing of stomata. **Dodecenyl succinic acid** also effects on stomatal closure.

Uses:

- Antitranspirants reduce the enormous loss of water by transpiration in crop plants.
- Useful for seedling transplantations in nurseries.

11.6.6 Guttation

During high humidity in the atmosphere, the rate of transpiration is much reduced. When plants absorb water in such a condition root pressure is developed due to excess water within the plant. Thus excess water exudates as liquid from the edges of the leaves and is called **guttation**. Example: Grasses, tomato, potato, brinjal and *Alocasia*. Guttation occurs through stomata like pores called **hydathodes** generally present in plants that grow in moist and shady places. Pores are present over a mass of loosely arranged cells with large intercellular spaces called **epithem** (Figure 11.18). This mass of tissue lies near vein endings (xylem and Phloem). The liquid coming out of hydathode is not pure water but a solution containing a number of dissolved substances.

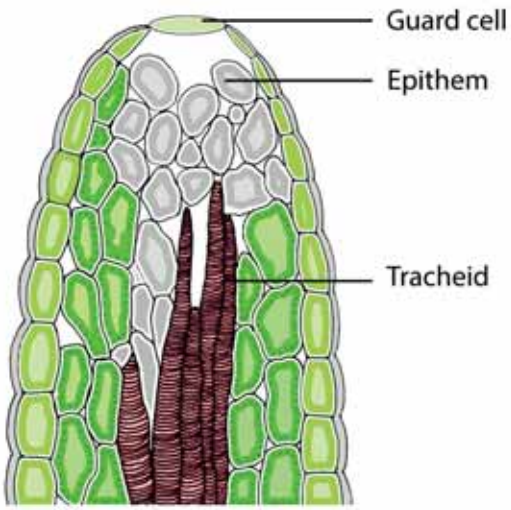


Figure 11.18: Structure of Hydathode

11.6.7 Measurement of Transpiration

1. Ganongs potometer

Ganongs potometer is used to measure the rate of transpiration indirectly. In this, the amount of water absorbed is measured and assumed that this amount is equal to the amount of water transpired.

Apparatus consists of a horizontal graduated tube which is bent in opposite directions at the ends. One bent end is wide and the other is narrow. A reservoir is fixed to the horizontal tube near the wider end. The reservoir has a stopcock to regulate water flow. The apparatus is filled with water from reservoir. A twig or a small plant is fixed to the wider arm through a split cock. The other bent end of the horizontal tube is dipped into a beaker containing coloured water. An air bubble is introduced into the graduated tube at the narrow end (Figure 11.19). keep this apparatus in bright sunlight and observe.As transpiration takes place, the air bubble will move towards the twig. The loss is compensated by water

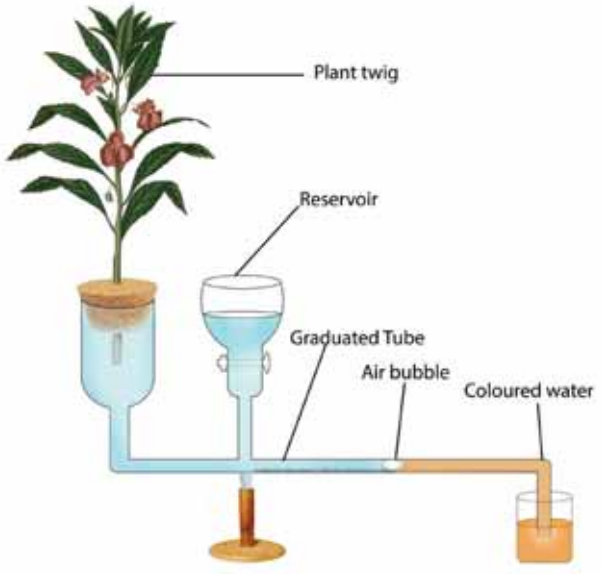


Figure 11.19: Ganongs Potometer